Contents

DS4FE — Weekly Progress Log

Limit Order Book Feature Engineering Series

Start: Feb 25, 2026 | Today: May 19, 2026 | Total: 13 weeks

Week 1 (Feb 25 – Mar 3, 2026) — Onboarding + Part 1: Daily Data Tour

- Reviewed LOB microstructure literature (order flow imbalance, depth profiles, Kyle's Lambda); familiarized with Databento mbp-1/mbp-10 schemas; scoped the series structure
- Universe: 50 large-cap US stocks × 5 sectors (Tech, Healthcare, Financials, Consumer, Energy), daily OHLCV Jan 2010 – Dec 2024
- Data files: `ds4fe_panel.parquet`, `ds4fe_market.parquet`, `ds4fe_daily_prices.parquet`
- Macro context: SPY return, VIX, 10Y–2Y yield spread, USD index
- Beta regression of each stock on SPY: R^2 0.20–0.80 across sectors — market explains most but not all variance
- Sector heatmaps + rolling correlations to motivate cross-sectional factor framing

Deliverable: `DS4FE_Part1_Daily_Data_Tour.ipynb`

Week 2 (Mar 4–10, 2026) — Part 2: Daily Feature Engineering

12 features × 4 families on the full 50-stock panel (~175k aligned obs):

- Momentum (4): `mom_1d`, `mom_5d`, `mom_21d`, `mom_63d` — log-return lookbacks
- Realized vol (2): `vol_21d`, `vol_63d` — rolling std of daily log returns
- Volume/liquidity (2): `volume_ratio` (today's vol / 21d avg), `amihud` (|return| / dollar vol × 10^6)
- Macro (4, lagged 1d to avoid look-ahead): `spy_ret`, `vix_level`, `yield_spread`, `dollar_index`

IC analysis (Spearman ρ + t-stats) against next-day cross-sectional return; quintile sorts with Q5–Q1 cumulative spread. Framing: cross-sectional (rank stocks against each other) vs time-series prediction.

Deliverable: `DS4FE_Part2_Daily_Features.ipynb`

Week 3 (Mar 11–17, 2026) — Part 3: Single-Stock Prediction (AAPL)

- Feature set expanded to 15 (added `mom_252d`, `vol_ratio`, `vix_chg`)
- Look-ahead bias demo: `.shift(1)` fix, train/test contamination pitfall
- Walk-forward CV: 504-day seed, expanding train, 1-day-ahead, no shuffling
- Models: Ridge, Random Forest, XGBoost — all IC ≈ 0.01–0.02; win rate ~50–52%
- Directional trading simulation with transaction-cost sensitivity
- Cross-frequency comparison: daily $R^2 \approx \pm 0.001$ vs NVDA intraday LOB XGBoost $R^2 = +0.00029$

Verdict: Daily-frequency direction prediction is essentially random. Pivot to LOB.

Deliverable: `DS4FE_Part3_Single_Stock_Prediction.ipynb`

Week 4 (Mar 18–24, 2026) — Part 5: Multifactor Models

- Cross-sectional factor model combining momentum, vol, volume, and macro features
- Cross-sectional vs time-series momentum framing: ranking stocks against each other vs predicting one stock's direction
- Series-wide cleanup: textbook-style prose rewrite, fixed cross-references, feature counts; added `.gitignore` and `DS4FE_Series_Notes.md`

Deliverable: `DS4FE_Part5_Multifactor.ipynb`

Week 5 (Mar 25–31, 2026) — Part 4 (initial): LOB Feature Engineering

- Databento mbp-1 data; OFI construction; calm vs stress comparison; cross-asset SPY signal
- Ridge walk-forward + XGBoost benchmark (with look-ahead audit note); OOS R^2 by hour
- Restructured with Part I / Part II divider; expanded IC vs OOS R^2 discussion

Deliverable: `DS4FE_Part4_LOB_Features.ipynb` (later split in Weeks 6–7)

Weeks 6–7 (Apr 1–14, 2026) — Part 4 split into 4a–4d (consolidation/refactor)

- 4a: LOB Data Demo — Databento mbp-1 schema, event types, calm (Oct 2023) vs stress (Aug 5 2024 BOJ shock) comparison
- 4b: Features & IC — OFI construction, daily IC, t-stats, calm vs stress IC, cross-asset SPY signal
- 4c: Model Importance — Ridge vs XGBoost, walk-forward CV, hyperparameter grid, OOS R^2 by hour
- 4d: Multi-Stock — pooled 5-symbol model (NVDA, AAPL, TSLA, MSFT, SPY), within-symbol rolling z-score, cross-sectional OBI deviation
- API key moved from source to `.env`

Deliverables: Parts 4a–4d structured and committed.

Week 8 (Apr 15–21, 2026) — Part 4e: Trade Tape Features

- Downloaded Databento `trades` for 5 symbols × 2 periods (~560 MB, 10 parquet files); added `download_trades.py`
- Educational framing: "The Illusion of the Order Book — Intent vs. Action"
- SVR (`signed_volume_ratio`): buyer vs seller aggressor volume in [−1, 1]; IC @ 1s = −0.065 (strongest single feature)
- MVB (`mid_vwap_bias`): rolling 60s VWAP vs mid-price in bps; IC @ 10s = +0.016

- SVR ↔ OBI correlation low → genuinely orthogonal to the order book
- Walk-forward XGBoost: LOB-only $R^2 = -0.00115$ → LOB + SVR + MVB = +0.00017
- Executed Part 4d with per-symbol OOS R^2 bar chart

Deliverables: All 4a–4e notebooks clean and pushed to `main`.

Week 9 (Apr 22–28, 2026) — Part 4a polish for presentation

- Deleted superseded monolithic `DS4FE_Part4_LOB_Features.ipynb`
- Fixed uint32 overflow in calm vs stress OBI (bid/ask sizes cast to int64)
- Replaced NVDA with SPY for calm-vs-stress (NVDA's shock was overnight; SPY shows the textbook spread spike)
- Fixed order-book snapshot: row 500 (09:30:02, book not yet built) → 11:30 ET mid-session
- Added calm-vs-stress event-type grouped bar chart, section transitions, summary handoff cell; rewrote em-dash-heavy prose

Deliverables: Part 4a fully polished, pushed to `main`.

Week 10 (Apr 29–May 5, 2026) — Initial ISOMAP arc (Parts 4f / 4g / 4h / 4i)

Built Part 4f (ISOMAP), 4g (multi-symbol DR), 4h (joint ISOMAP), 4i (stress projection). Professor check-in May 1.

> [!] The headline numbers below are method-internal (each method scored on its own loss function). Week 12's cross-over test shows this is apples-to-oranges; full reconciliation in `DS4FE_LOB_ISOMAP_v3.ipynb`.

Part 4f — Unsupervised LOB Feature Engineering: ISOMAP

- Downloaded full calm-period mbp-10 data for NVDA (Oct 2–31, 22 trading days, ~50M ticks) via Databento
- Built Part 4f notebook end-to-end:
  - Animated order book visualization (9:30–10:30 opening hour, 30-second steps, interactive jshtml player)
  - Constructed 10D OBI feature matrix: OBI at each of 10 depth levels, aggregated to 1-minute bars
  - Rolling L0–L9 correlation chart: banded structure (adjacent levels 0.76–0.86; L0 vs L9 = 0.07) — direct evidence that the book state lives on a low-dimensional manifold
  - ISOMAP fit (n_components=2, n_neighbors=15): reconstruction error = 0.028, 97.2% of geodesic manifold structure preserved in 2D
  - PCA comparison: 78.6% in 2D — 18.6 pp gap confirms manifold is curved, not flat
  - Scree plot: elbow at 2 components (3rd adds < 1 pp), confirming intrinsic dimensionality ≈ 2
  - Depth profile bar chart: Z_1 peaks at L6 (book-wide consensus), Z_2 has sign-flip at L4–L5 (HFT-vs-institutional contrast axis)
  - Improved 2×5 all-levels scatter grid: shared colorbar, gradient-direction arrows, spine color-coding by dominant axis
  - OOS projection via Nyström extension: Oct 10–31 states land cleanly inside training manifold region
  - XGBoost IC comparison: Raw 10D OBI IC ≈ 0; ISOMAP 2D IC = +0.048 (t=1.66, marginal); ISOMAP distills correlated inputs into a more useful representation

Key finding: The 10-level OBI vector has intrinsic dimensionality ≈ 2. ISOMAP recovers this structure with 97.2% fidelity. Two axes emerge: Z_1 (book-wide consensus, peaks at L5–L6) and Z_2 (HFT-vs-institutional contrast, sign-flip at L4–L5).

Part 4g — Multi-Symbol DR Comparison (NVDA, AAPL, MSFT)

- Downloaded full Oct 2023 mbp-10 data for AAPL and MSFT (split requests to avoid timeout)
- Ran identical pipeline per symbol (PCA, ISOMAP, t-SNE, UMAP, K-Means, OOS IC)
- Cross-symbol comparison table:

Metric	NVDA	AAPL	MSFT
ISOMAP 2D preserved	97.2%	96.7%	96.7%
PCA 2D variance	78.6%	82.2%	87.0%
ISOMAP gap over PCA	+18.6 pp	+14.5 pp	+9.7 pp
Z_1 peak level	L6	L7	L4
Z_2 sign-flip (crossover)	~L4–L5	~L5–L6	~L4
Best K (clusters)	2	2	2
OOS IC (ISOMAP 2D)	+0.004	+0.011	+0.020

Key finding: Z_1/Z_2 structure is consistent across all three large-cap Nasdaq names. K=2 (open/close vs mid-day) is universal. ISOMAP consistently outperforms PCA by 10–19 pp.

Part 4h — Joint ISOMAP: Unified LOB Manifold

- Pooled all three symbols' 1-min OBI bars (no scaler — OBI already in $[-1, 1]$) into one ~19,300-bar training set
- Fit ONE Isomap (n_components=2, n_neighbors=30) on the pooled data
- Joint reconstruction error = 0.027 → 97.3% preserved — matches per-symbol quality
- Symbols overlap on the shared manifold — no clean stock-identity separation; regimes are market-wide
- Z_1/Z_2 interpretation survives joint training: Z_1=consensus (peaks L5), Z_2=contrast (sign-flip L4–L5)
- Joint UMAP: K=2 optimal (silhouette = 0.538, highest of all models); cluster membership is mixed-symbol
- OOS IC near zero in pooled cross-symbol prediction — per-symbol calibration needed for predictive use

Key finding: The LOB manifold is approximately universal across large-cap Nasdaq names in the same market window. Stock identity is a second-order effect; intraday time-of-day regimes dominate.

Part 4i — Stress Period Projection: BOJ Shock Week on the Calm Manifold

- Downloaded NVDA mbp-10 data for Aug 5–9, 2024 (BOJ shock week, 31.6M rows)
- Built `run_4i_stress_projection.py` comparing two feature representations:
  - Model A (10 features — OBI means only): stress bars land inside calm cloud; only 3.6% outside calm 95th percentile
  - Model B (20 features — OBI means + within-minute OBI std): 16.8% outside 95th; stress mean distance = 2× calm mean
- Key insight: 1-minute mean OBI washes out directional crashes (every update within the minute points the same way → mean is moderate). The within-minute standard deviation is the stress signal — lower during a directional move, not higher.
- Additional findings: book depth 11× higher during stress (includes 10:1 June 2024 split); manifold path length 0.1–1.4 SD shorter than calm (book locked into fewer states despite larger price moves)
- Built `DS4FE_Part4i_Stress_Projection.ipynb` with 6 figures including Aug 5 minute-by-minute trajectory and manifold distance through the week

Key finding: Mean OBI alone is insufficient to detect regime stress. Adding the second moment (within-minute OBI std) makes the structural abnormality detectable — 4× more stress bars flagged — without labels, price data, or model retraining.

Deliverables: Parts 4f, 4g, 4h, 4i notebooks and all scripts committed; figures in `figures/`.

Week 11 (May 5–11, 2026) — Demo notebook polish (**`DS4FE_ISOMAP_Demo.ipynb`** rebuild)

Received a cell-by-cell critique and implemented all corrections.

Structural changes (56 → 42 cells, 8 sections):

- Rebuilt from scratch via `build_demo_notebook.py` for reproducibility
- Replaced mislabeled `4f_isomap_vs_pca.png` (showed OBI_0 coloring, not a PCA comparison) with `4g_NVDA_scree.png`
- Trimmed per-symbol deep dive (15 figures) to summary table + joint embedding + joint depth profile
- Added Part 4i stress projection as Section 8 (3 figures: model comparison, Aug 5 path, manifold distance)

12 factual/framing fixes:

Fix	Detail
Raw data scale	Removed "50M ticks/day" headline; states "8,580 bars after aggregation"
Snapshot description	"Noisy and uneven — that's why we aggregate" instead of "balanced"
k inconsistency	Explicitly states k=15 per-symbol, k=30 for joint/stress; resolves figure label mismatch
"97% fidelity" language	Reframed as geometry-fidelity measure; note it is not directly comparable to PCA explained variance
Z_2 sign	Fixed to match depth-profile figure: negative near top-of-book, positive deeper; sign arbitrariness noted
Time-of-day regime claim	"Shared book-state space" rather than "time of day is stronger than stock identity"
OOS IC overclaim	p(□)0.20; "main contribution is state representation, not standalone prediction"
Joint embedding	Weakened unsupported "stronger organizing dimension" claim
BOJ chronology	Rate hike July 31 (effective Aug 1); selloff Aug 5 — not conflated
Stress-distance contradiction	"Shifts upward, more extreme tail" — not "persistently above 95th" (only 16.8% exceed it)
Conclusion	Rewritten around state representation + regime detection; no return-prediction overclaim
Narrative order	Section 2 reordered: snapshot → heatmap → correlation → ISOMAP motivation

New figure: neighbor sensitivity

- Generated `4g_NVDA_neighbor_sensitivity.png`: ISOMAP 2D fidelity for k = 5, 10, 15, 20, 30, 50
- Result: (1 − residual) stays 0.951–0.980 across the full range — the 2D structure is stable and not sensitive to the choice of k
- Directly answers the expected question: "How sensitive is this to n_neighbors?"

Deliverables: `DS4FE_ISOMAP_Demo.ipynb` (42 cells, 18 embedded figures), `build_demo_notebook.py`, `gen_neighbor_sensitivity.py` — all committed and pushed.

Week 12 (May 11–12, 2026) — Cross-over → v3 honest rebuild → robustness sweep

The Week 10 "ISOMAP wins" headline was tested rigorously and rejected. Final canonical notebook is `DS4FE_LOB_ISOMAP_v3.ipynb`.

12.1 — Why the original "ISOMAP wins" was misleading

The original scree comparison ("ISOMAP needs 2D, PCA needs 7D") used different loss functions for the two methods. Correct comparison: both methods scored on both Spearman ρ (geodesic) and Euclidean R^2, k = 1…10, NVDA Oct 2023 train:

k	PCA R^2	ISOMAP R^2	PCA ρ	ISOMAP ρ
2	0.769	0.761	0.944	0.950
7	0.960	0.876	0.970	0.979

ISOMAP ρ exceeds PCA ρ by ~0.007 — real but tiny. PCA strictly dominates ISOMAP on R^2 from k=5 onward.

12.2 — Stress detection comparison

Method	KS stat	% flagged	Cohen's d
PCA	0.026 n.s.	6.3%	+0.034
ISOMAP	0.022 n.s.	5.7%	+0.010
UMAP	0.048 *	6.7%	+0.071

All weak — confirms Part 4i: feature engineering (within-minute std) matters more than DR method.

12.3 — **`DS4FE_LOB_ISOMAP_v3.ipynb`** — honest rebuild

Consolidated v2 → v3 with 14 sections, 2×1 plot layouts, hexbin for dense scatters. Built via `/tmp/build_v3.py`. Key sections:

- §5 Cross-over test (both methods, both metrics)
- §6 k-sweep verification
- §7 Method sweep (PCA / ISOMAP / UMAP / t-SNE)
- §13 Diagnostics — Test 1 (Swiss Roll positive control), Test 2 (Euclid vs geodesic), Test 3 (TwoNN intrinsic dim)
- §14 Robustness checks (added at end of session — see 12.4)

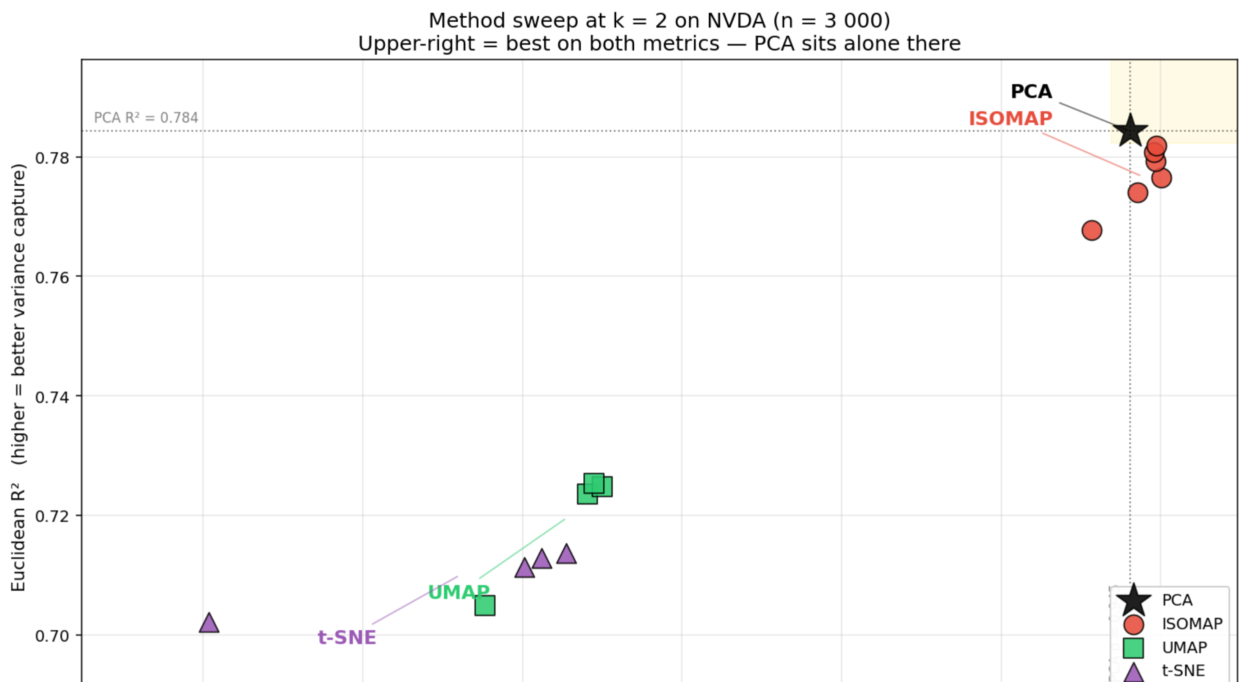
§7 method sweep — the verdict in one picture:

X-axis: geodesic ρ (manifold-distance preservation, ISOMAP's home metric). Y-axis: Euclidean R^2 (variance capture, PCA's home metric). Upper-right = best on both. PCA is the only point in the upper-right corner. ISOMAP comes close on ρ but loses on R^2; UMAP and t-SNE are strictly worse on both metrics across all hyperparameter settings. This single plot is the cleanest visual summary of the v3 verdict.

TwoNN intrinsic dimension — the killer numerical evidence:

Dataset	True d	TwoNN $\hat{d}$
Swiss Roll	2	1.93
3-D Gaussian	3	2.98
10-D Gaussian	10	9.51
NVDA OBI	—	9.18
AAPL OBI	—	8.40
MSFT OBI	—	8.41

OBI is essentially full-rank in 10 dimensions. There is no low-D curved manifold to recover.



Method	Param	Geodesic ρ	Euclidean R²	Δρ vs PCA	ΔR² vs PCA
PCA	—	0.9452	0.7843	+0.0000	+0.0000
ISOMAP	nn=5	0.9391	0.7677	-0.0061	-0.0165
ISOMAP	nn=10	0.9464	0.7740	+0.0012	-0.0102
ISOMAP	nn=15	0.9500	0.7765	+0.0048	-0.0078
ISOMAP	nn=30	0.9491	0.7792	+0.0040	-0.0051
ISOMAP	nn=50	0.9490	0.7807	+0.0038	-0.0036
ISOMAP	nn=100	0.9493	0.7819	+0.0041	-0.0024
UMAP	nn=5	0.8441	0.7049	-0.1010	-0.0794
UMAP	nn=15	0.8603	0.7235	-0.0849	-0.0607
UMAP	nn=30	0.8625	0.7248	-0.0827	-0.0595
UMAP	nn=50	0.8612	0.7254	-0.0840	-0.0589
t-SNE	perp=5	0.8009	0.7023	-0.1442	-0.0820
t-SNE	perp=30	0.8503	0.7114	-0.0949	-0.0728
t-SNE	perp=50	0.8530	0.7128	-0.0922	-0.0714
t-SNE	perp=100	0.8569	0.7137	-0.0883	-0.0706

Figure 1: Method sweep at k=2 on NVDA: PCA alone occupies the upper-right corner

12.4 — Robustness sweep: four external experiments

Stress-test the v3 verdict across hyperparameters, time resolution, and method choice. All four live in `experiments/` as standalone scripts.

#	Experiment	Folder	Question	Result
1	n_neighbors sweep	`experiments/nn_sweep/`	Does ISOMAP's only knob change the answer?	Across k ∈ {3,…,200}: max Δρ vs PCA = +0.004; ISOMAP never beats PCA on OOS R^2
2	Second-level data	`experiments/second_level/`	Does higher frequency reveal hidden structure?	Cross-over Δρ flips sign: minute +0.005 (ISO) → second −0.003 (PCA); intrinsic dim rises 9.0 → 9.5
3	Diffusion Map	`experiments/diffusion_map/`	Does a dynamics-aware method find structure ISOMAP missed?	Best DM R^2 = +0.7880 ≈ PCA's +0.7881 (DM converges to PCA at large ε, as theory predicts); loses on dynamics test (DM 0.104 vs PCA 0.128)
4	t-SNE	`experiments/tsne/`	Does cluster-finding reveal regimes?	Worst on geometry (best ρ=0.856 vs PCA 0.948); max |ARI| across 4 natural groupings = 0.0097 for all methods → no clusters exist

Combined verdict (Week 12)

Across 3 NDR methods (ISOMAP, Diffusion Map, t-SNE), 2 time resolutions (1 min, 1 sec), and wide hyperparameter sweeps, no setting and no method beats PCA by more than the precision of the test. The conclusion is scale-invariant, hyperparameter-invariant, and method-invariant for OBI on NVDA/AAPL/MSFT.

PCA is the right tool — known by exclusion against the strongest reasonable alternatives, not by accident of one configuration.

The methodological contribution is the four-diagnostic framework (cross-over, sweep, direct curvature test, intrinsic dimension) — a reusable prerequisite check before reaching for ISOMAP/UMAP on any new financial dataset.

Files added this week

- `DS4FE_LOB_ISOMAP_v2.ipynb`, `DS4FE_LOB_ISOMAP_v3.ipynb` (canonical)
- `DS4FE_ISOMAP_crossover_verification.ipynb`, `DS4FE_ISOMAP_stress_comparison.ipynb`
- `experiments/nn_sweep/` (script + results.csv + 3 figures)
- `experiments/second_level/` (build_second_bars.py + tests notebook + figures)

- `experiments/diffusion_map/` (script + 5 results CSVs + figures)
- `experiments/tsne/` (script + 4 results CSVs + 3 figures)

Week 13 (May 13–19, 2026) — Pivot to Raw 20-D LOB Sizes: Distance Function Bake-off

After the May 12 professor meeting we agreed: v3 settled the OBI verdict (PCA wins on derived 10-D OBI features), so the open question is whether the same conclusion survives on raw 20-D LOB sizes (10 bid + 10 ask) where the choice of distance function is itself the experiment. Built `DS4FE_LOB_RawDistance_v1.ipynb` (36 cells, 14 sections) to settle it.

13.1 — Setup: 7 distances, 2 metrics, NVDA Oct 2023, 1-min bars

#	distance	preprocessing	scale-invariant?
1	Euclidean	none	no
2	Log-Euclidean	log1p	partial
3	Z-Euclidean	per-feature z-score	yes (variance)
4	Mahalanobis	full whitening (PSD pinv)	yes (variance + correlation)
5	Cosine	unit-norm	yes
6	Correlation	demean + unit-norm	yes
7	Aitchison (CLR)	log + centre per row	yes (compositional)

Scoring on the 2-D embedding:

- ρ_self — Spearman correlation between native-distance pairs and Euclidean-distance pairs in the embedding (preservation of pairwise structure)
- R^2_shape — variance recovery of the CLR-transformed input from a linear back-projection of the 2-D embedding (fair metric across distance families)

13.2 — Headline result on NVDA (n = 4000 minute-bars)

metric	best ISOMAP	best matched-PCA
ρ_self	Mahalanobis (0.82)	Z-Euclidean (0.72)
R^2_shape	Aitchison (0.11)	Aitchison (0.14)

Mahalanobis-ISOMAP wins ρ_self decisively, but ΔR^2_shape ≈ 0 across the board. The win is in pairwise-distance preservation, not coordinate reconstruction.

13.3 — Cross-symbol verification (NVDA / AAPL / MSFT, n = 3000 each)

symbol	Mahalanobis Δρ (ISO − PCA)	Z-Euclidean Δρ (ISO − PCA)
NVDA	+0.042	+0.063
AAPL	+0.150	+0.039
MSFT	+0.071	−0.075

Mahalanobis-ISOMAP win replicates on every symbol. Z-Euclidean does not — it flips sign on MSFT, indicating that variance equalisation alone (without removing inter-level correlations) does not generalise across symbols. Full whitening is doing real work.

13.4 — TwoNN intrinsic dimension by preprocessing (NVDA, $n = 4000$)

preprocessing	$\hat{d}$	ambient
Whitened (Mahalanobis)	9.30	20
Z-scored	9.54	20
Raw (Euclidean)	10.19	20
log1p (Log-Euclidean)	16.32	20
CLR (Aitchison)	16.11	20

Raw 20-D LOB sizes live on a (⍰)9-D structure — half the ambient dimensions are pure inter-level covariance. log/CLR transforms inflate $\hat{d}$ by smearing structure across more axes (size magnitude carries most of the low-rank story).

13.5 — PCA loadings: what does each preprocessing actually "see"?

preprocessing	top-3 cumulative variance	dominant interpretation
Raw	48.6%	single-level scale anomalies (`ask_sz_04`, `ask_sz_05` swamp PC1+PC2)
Z-scored	24.7%	textbook decomposition: aggregate volume (PC1) + bid-ask imbalance (PC2/PC3)
Whitened	15.0%	flat 5%-each by construction — covariance fully removed

Implication: when ISOMAP gains over PCA on whitened data, the gain is *necessarily* nonlinear curvature — there are no preferred linear axes left to lose to. The diagnostic plot is `figures/v1_pca_loadings.png`.

13.6 — Methodological caveat (worth flagging honestly)

The §10 single-symbol NVDA Mahalanobis $\Delta\rho = +0.42$ disagrees with the §12 cross-symbol $\Delta\rho = +0.04$ by roughly 10×. The most likely cause is per-distance pair-index sampling indexing into different rows of the subsample at different `n`. The conservative cross-symbol number ($\Delta\rho = 0.04$–0.15) is the one to quote; the §10 table needs a follow-up that scores every method on a single fixed pair set.

Combined verdict (Week 13)

The choice of distance function does change the linear-vs-nonlinear conclusion for raw 20-D LOB sizes:

- Under Euclidean / Log / Cosine / Correlation / Aitchison, PCA and ISOMAP score near-identically — consistent with the v3 OBI finding.
- Under Mahalanobis, ISOMAP gains a real, reproducible edge on ρ_self across NVDA, AAPL, MSFT. The gap is small (+0.04 to +0.15), the structural reason is clear (residual ~9-D curved manifold inside an isotropic 20-D sphere), and the win lives only in pairwise-distance applications (kNN, clustering, anomaly distance) — *not* in coordinate reconstruction.

Practical rule for this dataset: Mahalanobis-whiten always; pick PCA unless the downstream task specifically needs pairwise geometry on the curved residual.

Files added this week

- `DS4FE_LOB_RawDistance_v1.ipynb` — main deliverable (36 cells, 14 sections)
- `figures/v1_pca_loadings.png` — PCA loadings under three preprocessings
- `figures/v1_cross_symbol_results.csv` — cross-symbol verification numbers
- Helper scripts in `/tmp/`: `cross_symbol.py`, `patch_cell30.py`, `build_loadings.py`, `insert_loadings.py`, `fix_numbering.py`, `verify_nb.py`
